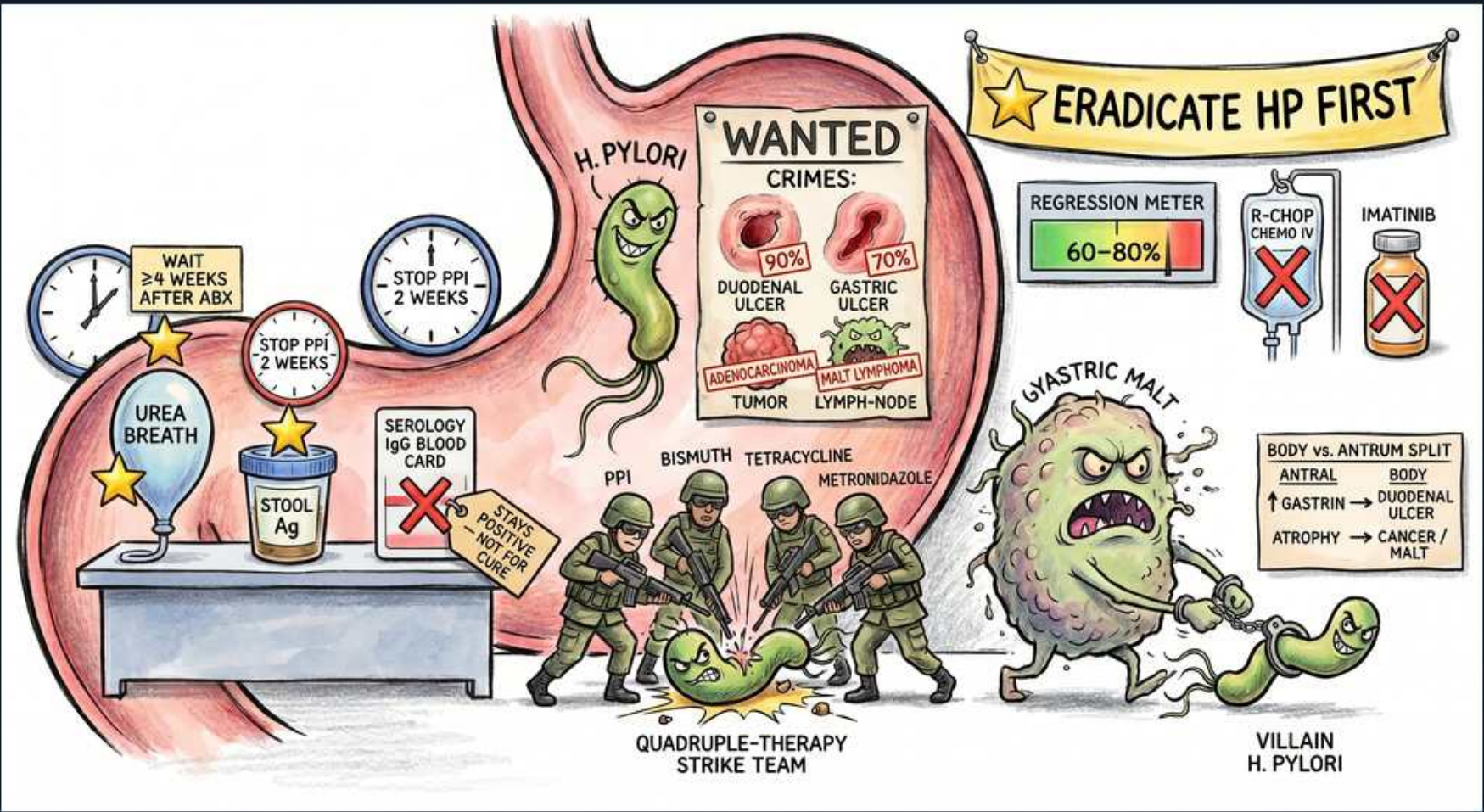


PUD — H. pylori: Diagnosis, Eradication & Gastric MALT



1. H. pylori villain

WHAT YOU SEE

A spiral bacterium with a flagellar tail glaring out, the central villain.

WHAT IT ENCODES

H. pylori is a gram-negative, urease-producing spiral organism that colonises gastric mucosa. Urease splits urea → ammonia, buffering acid (the basis of the urea breath test) and damaging mucosa. It is the dominant cause of peptic ulcers worldwide.

BOARD TRAP

'Most peptic ulcers are caused by stress/spicy food' → wrong; H. pylori and NSAIDs cause the vast majority.

2. WANTED poster — its crimes

WHAT YOU SEE

A WANTED poster listing crimes: duodenal ulcer 90%, gastric ulcer 70%, adenocarcinoma, MALT lymphoma.

WHAT IT ENCODES

H. pylori is found in ~90% of duodenal ulcers and ~70% of gastric ulcers, and is a WHO class I carcinogen — it causes gastric adenocarcinoma AND gastric MALT lymphoma. Eradication reduces ulcer recurrence and cancer risk.

BOARD TRAP

'H. pylori causes GERD / Barrett's / oesophageal adenocarcinoma' → wrong; if anything it's inversely associated. Its cancers are gastric (adenocarcinoma, MALT).

3. Urea breath test (gold star)

WHAT YOU SEE

A breath-test balloon labelled UREA BREATH wearing a gold star.

WHAT IT ENCODES

Urea breath test and stool antigen are the best NON-invasive active-infection tests and are the tests of CURE. Endoscopic biopsy/rapid urease is used when scope is already indicated.

BOARD TRAP

'Serology is the best test to confirm active infection' → wrong; antibodies persist after cure — use breath/stool for active disease and cure.

4. Stool antigen (gold star)

WHAT YOU SEE

A stool-antigen cup labelled STOOL Ag with a gold star.

WHAT IT ENCODES

Stool antigen detects active infection and confirms eradication. Like the breath test, it must be done off PPI and well after antibiotics to avoid false negatives.

BOARD TRAP

'A negative stool antigen taken on a PPI rules out H. pylori' → wrong; PPIs cause false negatives. Stop the PPI first.

5. Clock — wait ≥4 weeks after antibiotics

WHAT YOU SEE

A wall clock reading WAIT ≥4 WEEKS AFTER ABX.

WHAT IT ENCODES

Test of cure must be done at least 4 weeks after finishing antibiotics (and after bismuth), or you get false negatives from suppressed-but-not-eradicated bacteria.

BOARD TRAP

'Confirm cure immediately after the antibiotic course' → wrong; wait ≥4 weeks or you'll falsely call it cured.

6. Clock — stop PPI 2 weeks

WHAT YOU SEE

A second clock reading STOP PPI 2 WEEKS.

WHAT IT ENCODES

Hold the PPI for ~1-2 weeks before any active-infection test (breath, stool, biopsy urease) — acid suppression drives the organism into the body of the stomach and lowers its density, causing false negatives.

BOARD TRAP

'PPIs don't affect H. pylori test accuracy' → wrong; they cause false-negative active tests. Stop them before testing.

7. Red X on serology (IgG)

WHAT YOU SEE

A serology IgG blood card with a red X and tag STAYS POSITIVE — NOT FOR CURE.

WHAT IT ENCODES

IgG serology only shows past exposure; it stays positive for years after eradication, so it CANNOT confirm cure or reliably prove active infection.

BOARD TRAP

'Repeat serology to confirm eradication' → wrong; IgG remains positive. Use breath or stool antigen for cure.

8. Quadruple-therapy strike team

WHAT YOU SEE

Four soldiers labelled PPI, BISMUTH, TETRACYCLINE, METRONIDAZOLE crushing the villain.

WHAT IT ENCODES

Bismuth quadruple therapy (PPI + bismuth + tetracycline + metronidazole, ~14 days) is preferred first-line where clarithromycin resistance is high or after macrolide exposure. Clarithromycin triple therapy only where local resistance is low. Always confirm eradication afterwards.

BOARD TRAP

'A single antibiotic plus a PPI eradicates H. pylori' → wrong; monotherapy fails — combination (triple/quadruple) therapy is required.

9. Gastric MALT shrinking

WHAT YOU SEE

A lymph-node monster labelled GASTRIC MALT dissolving as the villain is dragged off in cuffs.

WHAT IT ENCODES

Gastric MALT lymphoma is H. pylori-driven. First-line treatment for early-stage, H. pylori-positive MALT is ERADICATION ALONE → regression in roughly 60-80%. (Label reads 'gyastric' — a Gemini typo for gastric.)

BOARD TRAP

'Gastric MALT lymphoma needs chemo/radiation up front' → wrong; eradicate H. pylori first in early HP-positive disease.

10. ERADICATE HP FIRST banner (gold star)

WHAT YOU SEE

A gold-star banner reading ERADICATE HP FIRST.

WHAT IT ENCODES

The single rule that ties the MALT story together: in H. pylori-positive, early/localised gastric MALT, eradication is the entire first-line treatment and often curative.

BOARD TRAP

'Always biopsy then jump to R-CHOP for any gastric lymphoma' → wrong for early HP+ MALT; eradicate first and reassess.

11. Regression meter 60-80%

WHAT YOU SEE

A green regression meter reading 60-80%.

WHAT IT ENCODES

Most early HP-positive MALT lymphomas regress with eradication. Non-responders, HP-negative cases, or those with t(11;18) translocation are less likely to respond and need radiation/immunochemotherapy.

BOARD TRAP

't(11;18)-positive MALT will still regress with antibiotics' → wrong; that translocation predicts antibiotic non-response.

12. Red X on R-CHOP & imatinib

WHAT YOU SEE

An R-CHOP chemo IV bag and an imatinib bottle, both with big red X marks.

WHAT IT ENCODES

R-CHOP is for diffuse large B-cell lymphoma, not early MALT. Imatinib (a tyrosine-kinase inhibitor for CML/GIST) is a pure distractor here.

BOARD TRAP

'Imatinib treats gastric MALT lymphoma' → wrong; it's a distractor. Early MALT = eradicate H. pylori.

13. Antrum vs body split

WHAT YOU SEE

A small card: ANTRAL → ↑gastrin → duodenal ulcer; BODY → atrophy → cancer / MALT.

WHAT IT ENCODES

Pattern matters: antral-predominant gastritis raises gastrin → increased acid → DUODENAL ulcers; body/corpus-predominant gastritis causes atrophy → reduced acid → higher risk of gastric cancer and MALT.

BOARD TRAP

'H. pylori always raises cancer risk by raising acid' → wrong; it's the body-predominant ATROPHIC (low-acid) pattern that drives gastric cancer/MALT.